Vector Functions

15 September 2025

Thus far, we have worked with **functions**—a function f is a rule that assigns each element, say x, in a *domain* D to a single value $f(x)$. We now turn to functions that take a real number and, to each number, assign a corresponding vector.

A **vector function** $\vec{r}$ assigns every number t in a domain to a unique vector $\vec{r}(t)$:

$$\vec{r}(t) = \begin{pmatrix} f(t) \\ g(t) \\ h(t) \end{pmatrix} = f(t)\,\vec{i} + g(t)\,\vec{j} + h(t)\,\vec{k}.$$

The functions f, g, and h are called the **component functions** of $\vec{r}$. As the value of t varies, the points

$$(x, y, z) = (f(t), g(t), h(t))$$

form a **curve** in space.

Example. Let $\vec{r}(t) = t^3\,\vec{i} + \ln(3-t)\,\vec{j} + \sqrt{t}\,\vec{k}$. Then,

$$f(t) = t^3, \quad g(t) = \ln(3-t), \quad h(t) = \sqrt{t}.$$

The domain of $\vec{r}$ consists of the values of t for which f, g, and h are defined: in this case, $3 - t > 0$ and $t \geq 0$; hence, the domain of $\vec{r}$ is the interval $[0, 3)$.

Example. Consider the curve defined by the vector function

$$\vec{r}(t) = \begin{pmatrix} a + pt \\ b + qt \\ c + rt \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} + t \begin{pmatrix} p \\ q \\ r \end{pmatrix}, \quad t \in \mathbb{R}.$$

We have seen this before: $\vec{r}$ is the equation of a line that passes through the point (a, b, c) and is parallel to the vector $p\,\vec{i} + q\,\vec{j} + r\,\vec{k}$. Note that if we limit t to an interval (rather than all of $\mathbb{R}$), we obtain a line segment.

Example. Consider the curve whose vector equation is

$$\vec{r}(t) = \cos t\,\vec{i} + \sin t\,\vec{j} + t\,\vec{k}.$$

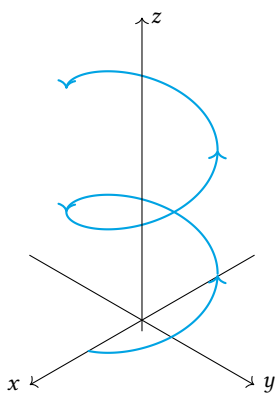
The corresponding parametric equations are

$$x = \cos t, \quad y = \sin t, \quad z = t.$$

Observe that

$$x^2 + y^2 = \cos^2 t + \sin^2 t = 1,$$

and so, the curve $\vec{r}(t)$ must be traveling counterclockwise along the path of the unit circle. Since $z = t$, $\vec{r}(t)$ moves upwards in the direction of z as t increases. We end up with a spiral—a curve known as a *helix*.



Calculus on Vector Functions

Let $\vec{r}(t)$ be a vector function given by

$$\vec{r}(t) = f(t)\,\vec{i} + g(t)\,\vec{j} + h(t)\,\vec{k}.$$

We define the **limit** of a vector function by taking the limits of its component functions:

$$\lim_{t\to a} \vec{r}(t) = \lim_{t\to a} f(t)\,\vec{i} + \lim_{t\to a} g(t)\,\vec{j} + \lim_{t\to a} h(t)\,\vec{k}.$$

From this definition, we can thus compute the **derivative** of a vector function component-wise:

$$\vec{r}'(t) = f'(t)\,\vec{i} + g'(t)\,\vec{j} + h'(t)\,\vec{k}.$$

Note that the derivative is a *vector*: in particular, $\vec{r}'(t)$, when evaluated at a point, is the **tangent vector** to the curve defined by $\vec{r}$ at the given point. Oftentimes, it is helpful to determine the **unit tangent vector**, given by

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Remark. We use the independent variable t since, in many applications, vector functions often describe how a particle moves in space with respect to *time*. The first and second derivatives of $\vec{r}$ thus gain a physical meaning: if $\vec{r}(t)$ gives a position vector of particle in space at time t, we can think of $\vec{r}'(t)$ as a velocity vector, and $\vec{r}''(t)$ as an acceleration vector.

Example. Let us find parametric equations for the tangent line to the helix

$$\vec{r}(t) = 2\cos t\,\vec{i} + \sin t\,\vec{j} + t\,\vec{k} \text{ when } t = \pi/2.$$

A line is determined by a point on the line and a direction vector. Since the line is tangent to the curve at $t = \pi/2$, it must pass through the point

$$\vec{r}\left(\frac{\pi}{2}\right) = 2\cos\frac{\pi}{2}\,\vec{i} + \sin\frac{\pi}{2}\,\vec{j} + \frac{\pi}{2}\,\vec{k},$$

which is the point $P(0, 1, \pi/2)$. To determine a direction vector, note that the tangent line must be parallel to the tangent vector: we thus compute the first derivative of $\vec{r}$:

$$\vec{r}'(t) = -2\sin t\,\vec{i} + \cos t\,\vec{j} + \vec{k}.$$

When $t = \pi/2$, the tangent vector at that point is

$$\vec{r}'\left(\frac{\pi}{2}\right) = -2\sin\frac{\pi}{2}\,\vec{i} + \cos\frac{\pi}{2}\,\vec{j} + \vec{k} = -2\,\vec{i} + 0\,\vec{j} + \vec{k}.$$

Thus, the tangent line through the point $(0, 1, \pi/2)$ is parallel to the vector $-2\,\vec{i} + 0\,\vec{j} + \vec{k}$. Hence, the vector equation of our line is given by

$$\vec{r} = \begin{pmatrix} 0 \\ 1 \\ \pi/2 \end{pmatrix} + t\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Our parametric equations are thus

$$x = -2t, \quad y = 1, \quad z = \frac{\pi}{2} + t.$$

Many familiar rules and properties of differentiation can be extended to vector functions.

Proposition 1 (Differentiation Rules)**.** *Let $\vec{u}(t)$ and $\vec{v}(t)$ be differentiable vector functions, let c be a scalar, and let $f(t)$ be a real-valued function. Then,*

1. *Linearity:* $\frac{d}{dt}[\vec{u}(t)+\vec{v}(t)] = \vec{u}'(t)+\vec{v}'(t)$ *and* $\frac{d}{dt}[c\vec{u}(t)] = c\vec{u}'(t)$
2. *Product Rule:* $\frac{d}{dt}[f(t)\vec{u}(t)] = f'(t)\vec{u}(t)+f(t)\vec{u}'(t)$
3. *Dot Product:* $\frac{d}{dt}[\vec{u}(t)\cdot\vec{v}(t)] = \vec{u}'(t)\cdot\vec{v}(t)+\vec{u}(t)\cdot\vec{v}'(t)$
4. *Cross Product:* $\frac{d}{dt}[\vec{u}(t)\times\vec{v}(t)] = \vec{u}'(t)\times\vec{v}(t)+\vec{u}(t)\times\vec{v}'(t)$
5. *Chain Rule:* $\frac{d}{dt}[\vec{u}(f(t))] = \vec{u}'(f(t))f'(t)$

As with curves on the plane, we can compute the lengths of curves in space. The **arc length** of a curve defined by a vector function $\vec{r}(t)$ from $t=a$ to $t=b$ is given by

$$L = \int_a^b |\vec{r}'(t)|\,dt.$$

Example. Let us find the length of the arc of the helix

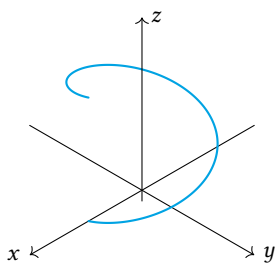
$$\vec{r}(t) = \cos t\,\vec{i}+\sin t\,\vec{j}+t\,\vec{k}$$

from the point $(1,0,0)$ to the point $(1,0,2\pi)$. The tangent vector is

$$\vec{r}'(t) = -\sin t\,\vec{i}+\cos t\,\vec{j}+\vec{k},$$

and we can thus compute its length to be

$$|\vec{r}'(t)| = \sqrt{(-\sin t)^2+(\cos t)^2+1^2} = \sqrt{2}.$$



The arc from $(1,0,0)$ to $(1,0,2\pi)$ is described by the parameter interval $0 \le t \le 2\pi$. Thus,

$$L = \int_0^{2\pi} \sqrt{2}dt = 2\sqrt{2}\pi.$$